

電腦對局理論 Homework 2

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December 6, 2025

1 Implementation detail

The following is the design of the code submission version solver.

1.1 MCTS iteration

The MCTS iteration run as below:

```
procedure MCTSITERATION
  PV  $\leftarrow$  Get_PV()
  if PV.state == Termination then
    BackPropagation(PV.leaf, TerminationState, num_simulate)
    AmafBackPropagation(PV.leaf, TerminationState, num_simulate)
  return
  expand(PV.leaf)
  avg_score  $\leftarrow$  0
  sim_records  $\leftarrow$  {}
  for  $child \in Children(PV.leaf)$  do
    node_score, record  $\leftarrow$  RandomSimulate(child.pos, N_sim_leaf)
    sim_records = sim_record  $\cup$  record
    BackPropagation(child, node_score, N_sim_leaf)
    avg_scores += (-node_score)
  avg_scores  $\leftarrow$   $\frac{avg\_score}{Children(PV.leaf).size()}$ 
  N_amaf  $\leftarrow$  UpdateAmafLeaves(PV.leaf, avg_score, sim_records)
  AmafBackPropagation(PV.leaf, avg_scores, sim_records)
  cut_node = PV.leaf
  while cut_node  $\neq$  root do
    ProgressivePruning(cur_node)
    cut_node  $\leftarrow$  cut_node.parent
```

The agent maintain two types of nodes, calling Nodes and AMAF_Nodes, the structure of two type of nodes are identical, only the values in the nodes are differ, and therefore we can apply the same updating function to both.

RamdonSimulate(pos, N) will perform N simulations on pos, and return its average and record the moves happened at the simulation.

N_sim_leaf is the number of simulations at the leaf, BackPropagation will assume we perform N_sim_leaf simulations, and obtain the return value node_score each times, then record the solution to the PV. Which will decrease the variance, but the performance is not change too much in total.

AmafBackPropagation is identical with BackPropagation, but only update the AMAF nodes. The UpdateAMafLeaves function will update all AMAF nodes of the children w.r.t PV.leaf.

In the following parts, I use a human designed PieceValue Table as below

將	士	象	車	馬	包	卒
7	8	6	5	3	9	1

Pieces Value

1.2 Simulation and improvement

For each single MCTS iteration, the agent will perform 10 simulations.

```

1: procedure RANDOMSIMULATE(pos)
2:   while pos is not terminated do
3:      $Moves \leftarrow gen\_move(pos)$ 
4:     for  $mv \in Moves$  do
5:        $W_{mv} \leftarrow MoveEvaluate(mv)$ 
6:        $next \leftarrow sample(Moves)$   $\triangleright$  w.r.t  $P(next = mv) = \frac{W_{mv}}{\sum_{m \in Moves} W_m}$ 
7:       pos.do_move(next)
8:   return TerminationBoardEvaluate(pos)

```

In the simulation, the next move is randomly selected with respect to the distribution of $MoveEvaluate$. This operation make it slower about 10 times, but largely improve the simulation quality.

$basic_move_value$ and C are constants, which are setting as 10 and 100 respectively.

At initial experiment, rather than the MCS model of 1000 simulations for each single move, the MCTS model with only 100 simulations totally can win the MCS model stably. Therefore the modification is worthy.

1.4 RAVE

The Rapid Action Value Estimation method used at the leaf. I use the following function to update the AMAF leaves, calculate the equivalent times of simulation in AMAF tree and return.

```
procedure UPDATEAMAFLEAVES(leaf, avg_score, sim_record)
  AMAF_N_simulate  $\leftarrow$  0
  for child  $\in$  Children(leaf) do
    if leaf.Count(child.move) > 0 then
      update_node(child, avg_score, leaf.Count(child.move))
      AMAF_N_simulate += leaf.Count(child.move)
  return AMAF_N_simulate
```

After this, the solver will viewed as PV.leaf performed AMAF_N_simulate simulations and obtain score *avg_score* each times, and update the AMAF tree w.r.t this.

1.5 Progressive Pruning

The Progressive Pruning is used at PV in each iterations.

```
procedure PROGRESSIVEPRUNING(node)
   $score_{min} = node.mean - r \times node.std$ 
  for child  $\in$  Children(node) do
     $score_{child} = child.mean + r \times child.std$ 
    if  $score_{child} < score_{min}$  then
      if child.activated then
        deactivate(child)
    else
      if child.deactivated then
        activate(child)
```

The pruning is performed with respect to the original MCTS value, not AMAF or RAVE value, and the pruning is happened after the simulation and BackPropagation ended.

2 Experiments